

Travel Insurance Profit Optimization for Allianz Partners

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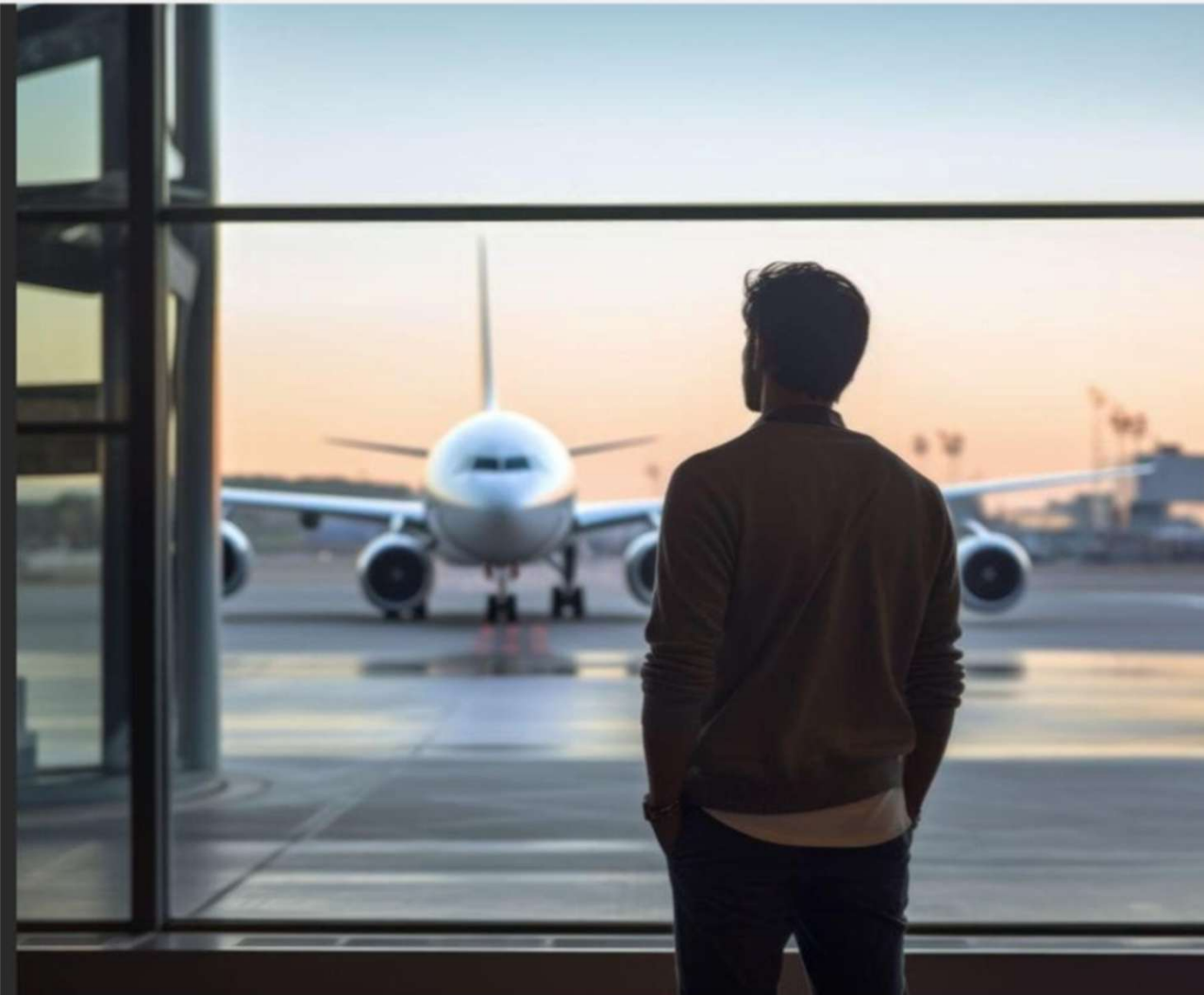


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01

Business Overview





Introduction

Flight delays can lead to high insurance claim payouts, creating uncertainty for providers.

This project aims to help an insurance company (Allianz Partners) improve decision-making and profitability by leveraging historical U.S. flight delay data (2022–2024).

Our solution uses predictive analytics to manage flight delay likelihood, optimize pricing, and enhance customer experience.

Business Questions

01

Forecast

What is the projected number of high-delay flights in 2025, and what is the expected cost of insurance claims payout associated with these delays?

02

Optimize

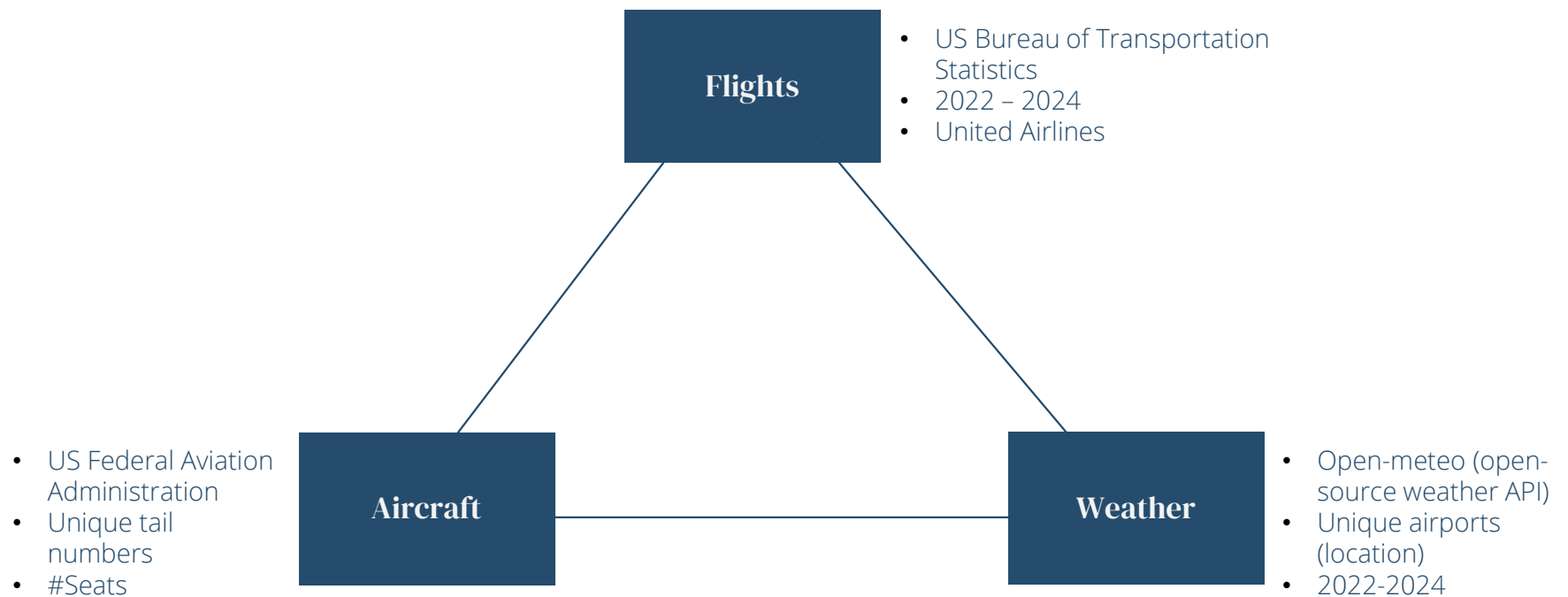
What is the optimal premium price to maximize profit in 2025, and will three differentiated premium tiers yield higher profitability compared to a single flat-rate premium?

03

Segment

How can the corporation predict whether a future flight will be delayed and adjust the insurance pricing accordingly?

Data Preparation



An aerial photograph of an airport tarmac. Three white commercial airplanes are parked in a row, each with a yellow ground support vehicle (GSE) nearby. The tarmac is marked with orange and white lines. A white rectangular box is superimposed over the middle airplane and the text 'Exploratory Data Analysis'.

Exploratory Data Analysis

02

United Airlines Domestic Network Overview



YEAR

☐ 2022

☐ 2023

☒ 2024



OVERVIEW



TREND



3.83M

Flight Arrivals

CY: 1,336,509 ▲ 5.1%
PY: 1,272,192



81.3%

OnTime %

CY: 81.4% ▲ 0.7%
PY: 80.7%



253

Airports

CY: 221 ▼ 0.9%
PY: 223



251

Cities

CY: 220 ▼ 0.9%
PY: 222



1,709

Flight Routes

CY: 1,448 ▲ 2.1%
PY: 1,418

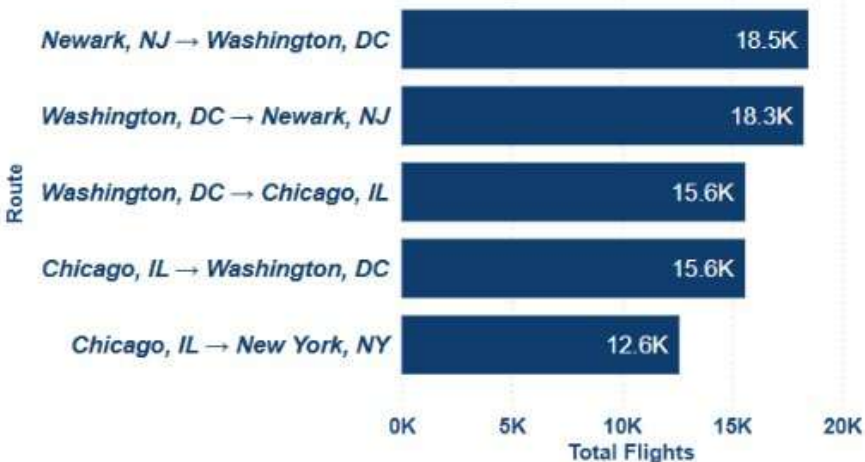


411M

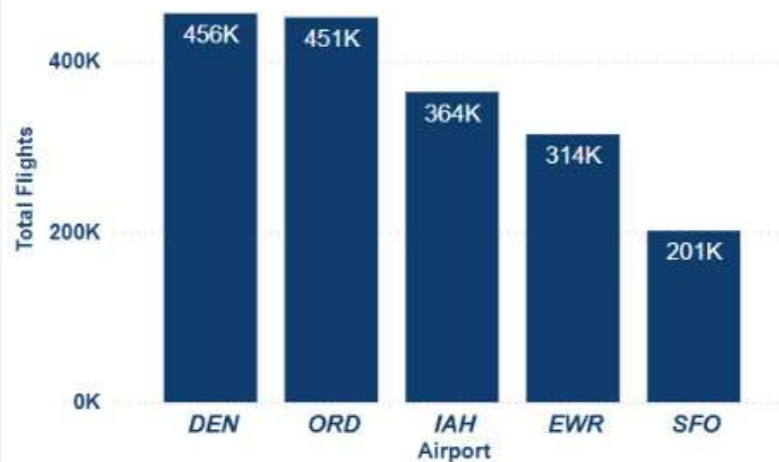
Passengers

CY: 144,000,354 ▲ 3.6%
PY: 138,976,556

Top 5 Flight Routes



Top 5 Busiest Airports





03 Forecasting

Technical Overview



Flight Delay

Low Delay: ≤ 60 minutes
Moderate Delay: $= 61-359$ minutes
High Delay: ≥ 6 hours (claim-eligible)

Model Used: SARIMA (captures monthly trend and seasonality)



Cost of Insurance Claims

Passenger Load Factor: 83.53%*
Insurance Uptake: 70%**
Insurance Payout: \$300 per valid claim***

What is the projected number of high-delay flights in 2025, and what is the expected cost of insurance claims payout associated with these delays?

*Passenger Load Factor based on Statista 2023

**Insurance Uptake based on Insubuy 2024 statistics

***Insurance Payout amount based on Allianz OneTrip Basic Plan

What is the projected number of delayed flights (high) in 2025?



HISTORICAL FLIGHT DELAY*

2024: 4,922
Annual Average: 4,657



2025 FLIGHT DELAY FORECAST*

5,227
 ↑ 6.20% vs 2024
 ↑ 12.23% vs Annual Average

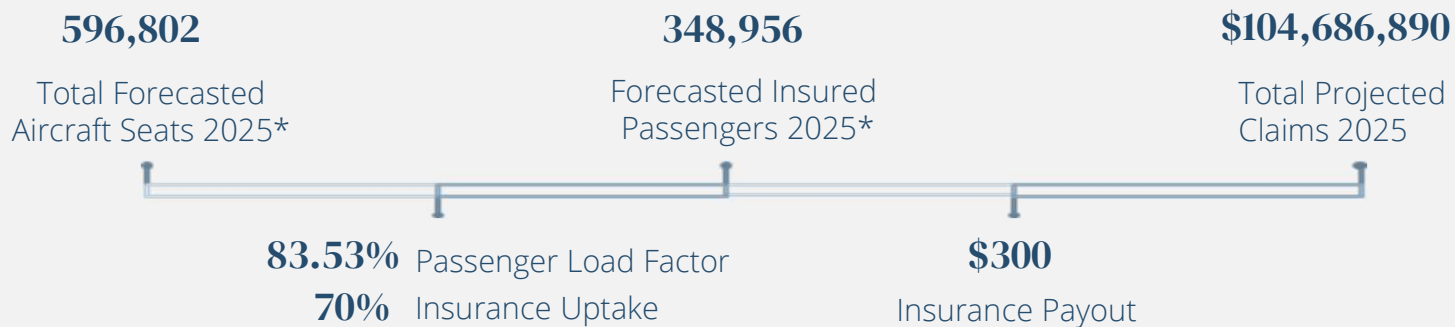
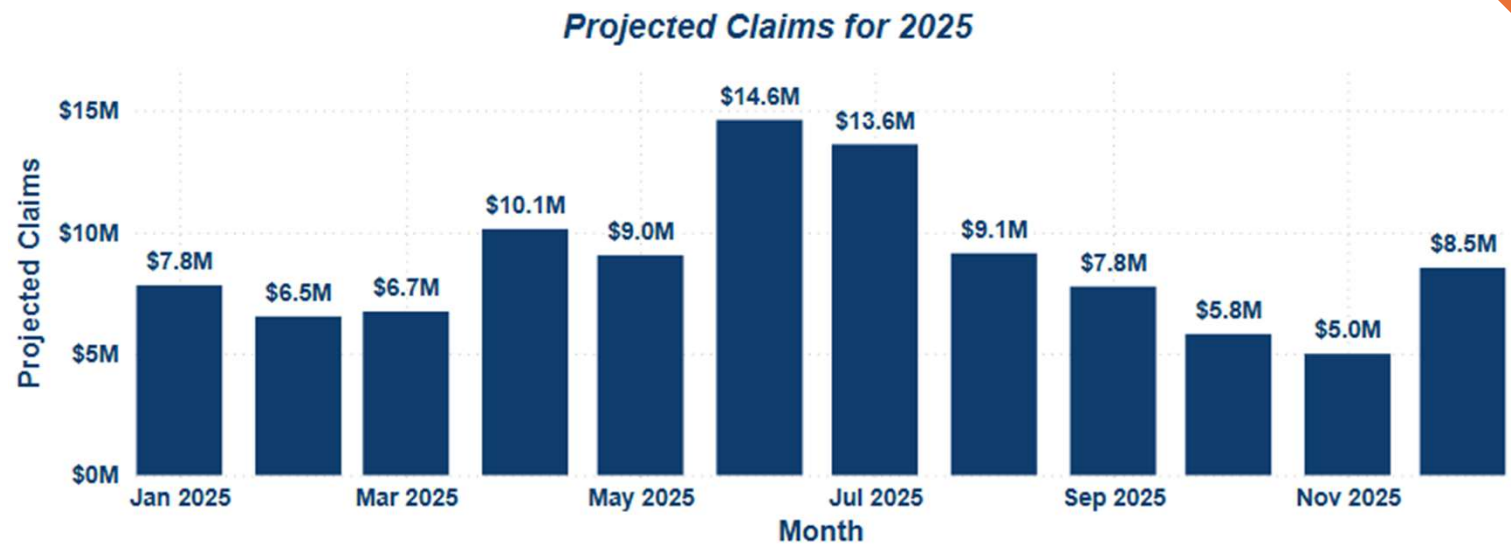


METRICS

MAE: 14.49
RMSE: 18.15
MAPE: 3.75%

*High Delay

What is the expected cost of insurance claims payout associated with the (high) delays?



*High Delay



Technical Overview



Linear Regression Model

$$Y = -0.8245x + 0.0165$$

Y: Change in Insurance Buyer Volume (%)

X: Price Variation (%)

$$R^2 = 97.89\%$$



Price Optimization

Objective: Maximize Profit

Variable: Price variation (%)

Constraints: Price Variation in High Delay $\leq 30\%$

What is the optimal premium price to maximize profit in 2025, and will three differentiated premium tiers yield higher profitability compared to a single flat-rate premium?

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Single Flat

-9.6800%



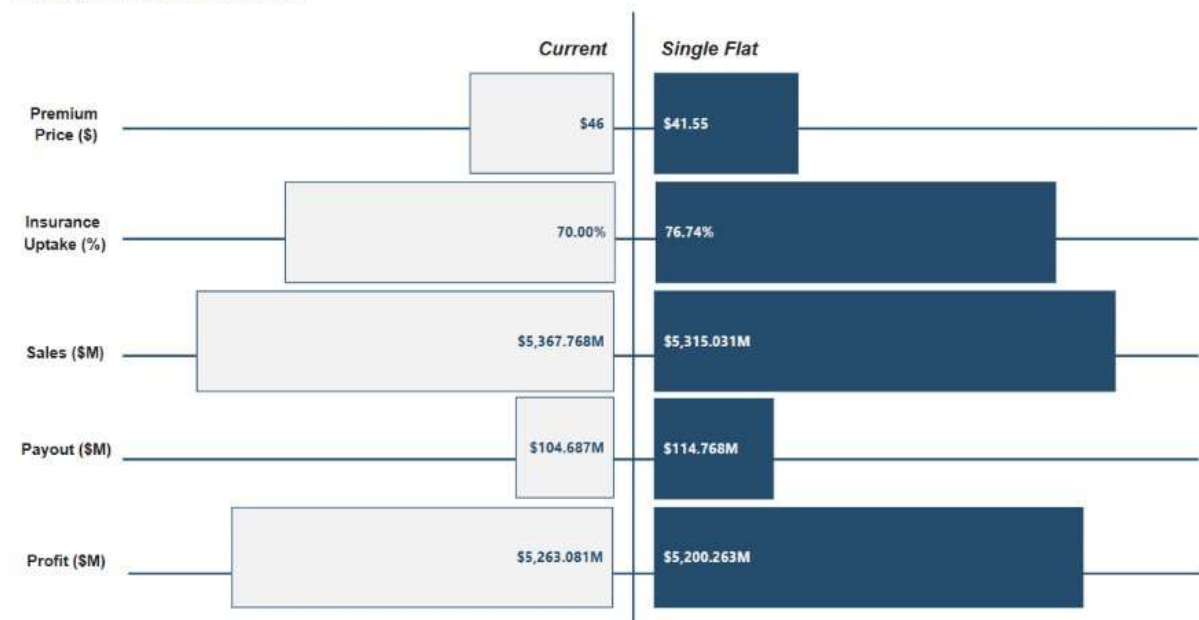
Current vs Single

Single vs 3 Tiers

Profit Optimization

Allianz 

Pricing Impact Comparison



Results



Low Delay Premium:
\$51.35 (+11.64%)



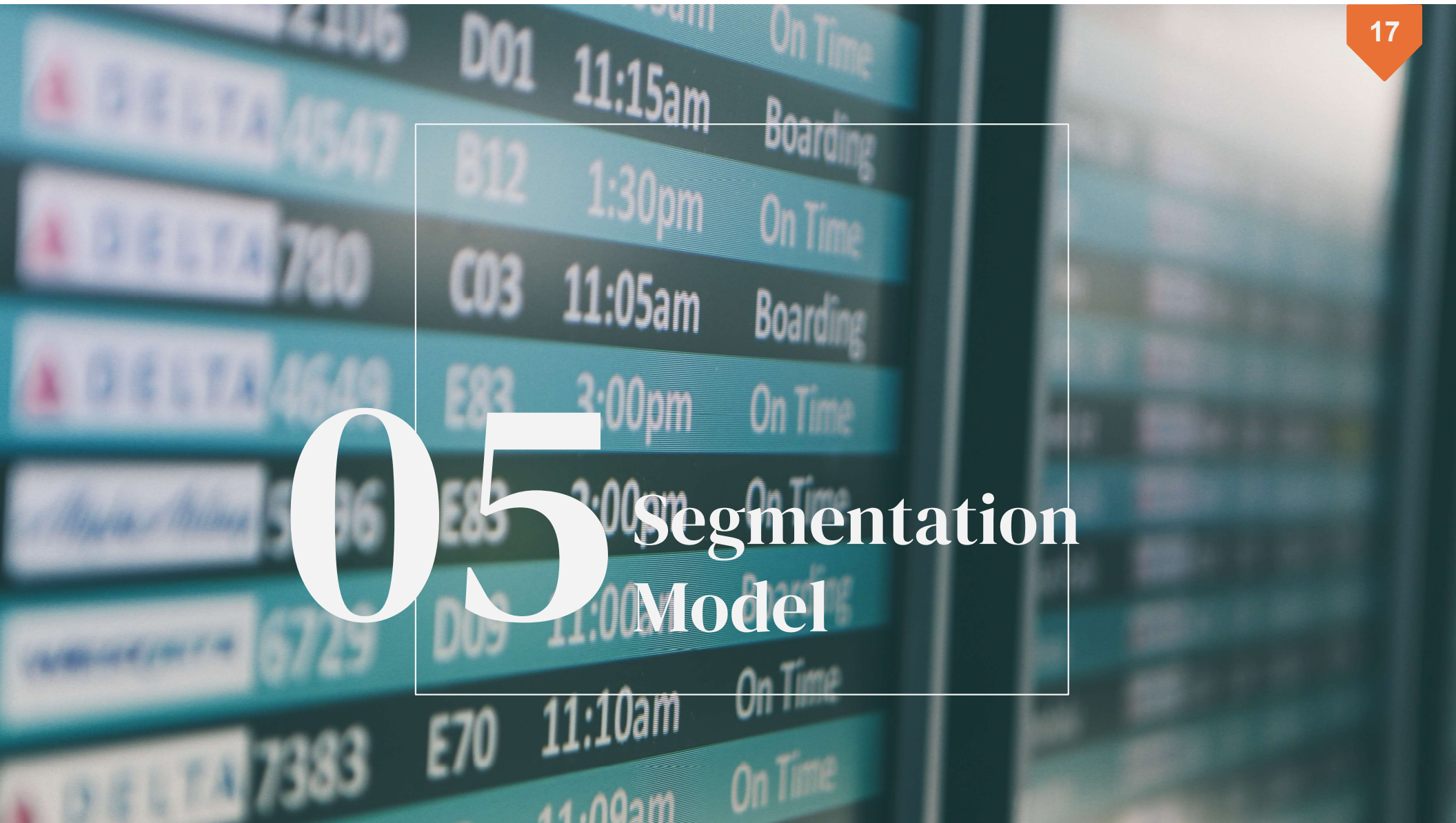
Moderate Delay Premium:
\$51.35 (+11.64%)



High Delay Premium: \$59.8
(+30%)

What is the optimal premium price to maximize profit in 2025, and will three differentiated premium tiers yield higher profitability compared to a single flat-rate premium?

05 Segmentation Model



Segmentation Model

Feature Engineering

- Select features
- Reduce cardinality
- Create new features



Model Selection

- Random forest classifier
- Logistic regression
- Gradient boosting
- Multinomial NB

Model Training

- Dataset balancing
- Increase sample size



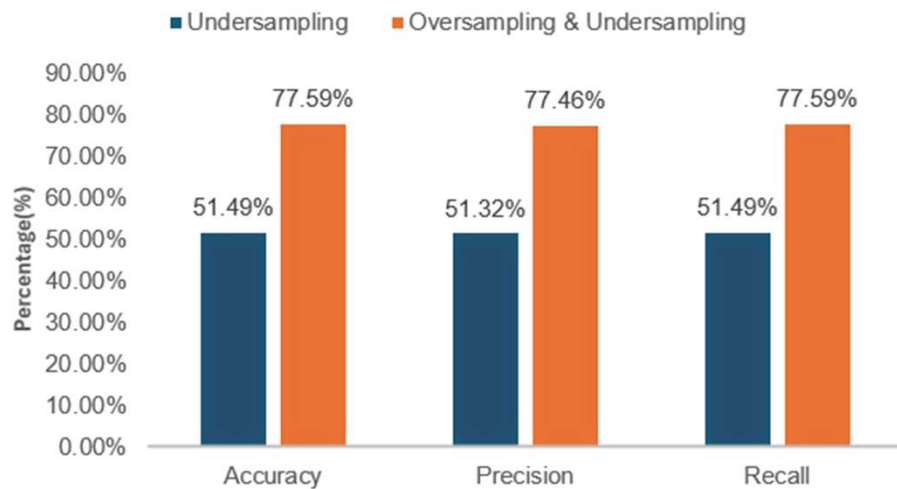
Model Tuning

- Grid search cross validation
- Random search cross validation

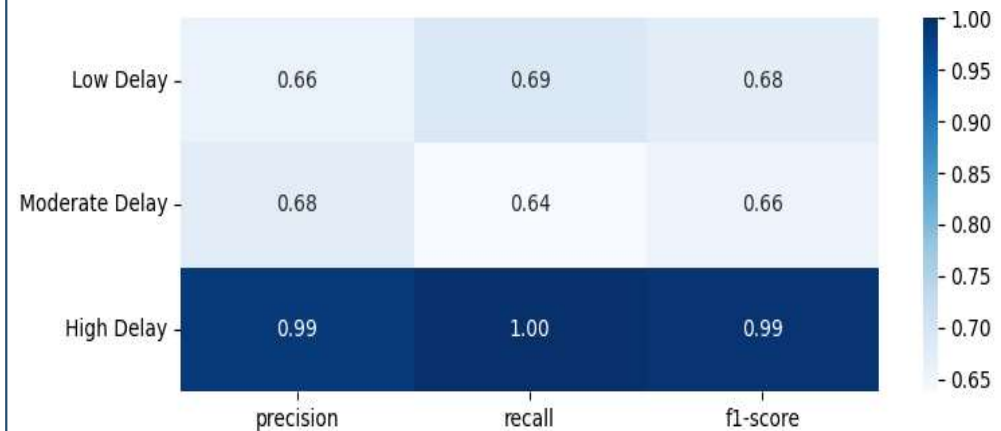
How can the corporation predict whether a future flight will be delayed and adjust the insurance pricing accordingly?

Random Forest Classifier

EVALUATION METRICS



CLASSIFICATION REPORT BY DELAY CATEGORY



How can the corporation predict whether a future flight will be delayed and adjust the insurance pricing accordingly?

A photograph of an airplane at sunset. The sky is filled with orange and red clouds. The airplane's wing and engine are visible in the foreground. The text 'Final Insights' is overlaid in white serif font.

Final
Insights

06

Conclusion



Forecasting



United Airlines is forecasted to have **5,227** high-delay flights and Allianz will spend **\$104 MM** on insurance claims of delayed flights in 2025.

Optimization



By increasing the insurance premium to \$59.8 (low), \$51.35 (moderate) and \$51.35 (high), Allianz will be able to **increase profit** from \$5.26B to **\$5.44B** in 2025.

Segmentation



With access to private data sources and greater computing power, Allianz will be able to **improve classification accuracy** and achieve the goal of optimized profit.



Thank you!

Any Question?

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